

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Summer 2024

Course: B. Tech.

Branch: Civil Engineering

Semester: VI

Subject Code & Name: BTCVPE604D Water Power Engineering

Max Marks: 60

Date: 21/06/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1	Solve Any Two of the following.		12
A)	Discuss the significance of hydro power as a source of energy.	CO1	6
B)	Identify the main components of a hydro power project.	CO1	6
C)	Define load curve and load duration curve. Why are they important in power generation?	CO1	6
Q.2	Solve Any Two of the following.		12
A)	Explain the factors involved in the hydraulic design of pen-stocks.	CO1	6
B)	Discuss the importance of proper tunnel lining.	CO1	6
C)	Describe the design criteria for a simple surge tank.	CO1	6
Q. 3	Solve Any Two of the following.		12
A)	Illustrate the general arrangement of a typical hydro power station.	CO2	6
B)	Describe the sub-structure and superstructure of a power house.	CO2	6
C)	Discuss the economic considerations in the development of underground power stations.	CO2	6
Q.4	Solve Any Two of the following.		12
A)	What factors influence the choice of a specific type of turbine?	CO2	6
B)	What is cavitation and how does it affect turbine performance?	CO2	6
C)	What are the functions of a tail race in a hydro power plant?	CO2	6
Q. 5	Solve Any Two of the following.		12
A)	What is the purpose and General Layout of a pumped storage plant?	CO3	6
B)	Discuss the economic benefits of pumped storage plants.	CO3	6
C)	Describe the advantages and limitations of tidal power stations.	CO3	6

*** End ***